

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – I

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Class: TYIT	Batch: 01
Date of Assignment: 24/07/2026	Date/Time of Submission: 24/07/2026

Part A - Theory:

a) Connect to a CSV Dataset and Explore the Tableau Interface

Tableau allows users to connect to CSV files for data analysis. After loading the dataset, users can explore the Data pane, Sheets, Rows, Columns, Marks card, and Toolbar to create visualizations efficiently.

b) Identify and Differentiate Measures and Dimensions

Dimensions are qualitative fields (e.g., Name, Category, Region) used for grouping and categorizing data. Measures are quantitative fields (e.g., Sales, Profit, Quantity) used for calculations and analysis.

c) Convert Fields Between Discrete and Continuous

In Tableau, fields can be displayed as **Discrete (blue)** or **Continuous (green)**. Discrete fields create headers and separate categories, while continuous fields create axes for numerical analysis, changing the visualization style.

d) Create and Customize a Basic Bar Chart

A bar chart is used to compare values across different categories. Users can customize the chart by changing colors, sorting data, adding labels, adjusting sizes, and formatting axes for better readability.

e) Build a Line Chart for Time-Series Analysis

A line chart displays trends over time by connecting data points with lines. It is useful for analyzing changes, patterns, and growth across dates or time periods.

f) Create a Geographic Map Visualization

Tableau automatically recognizes geographic fields such as Country, State, or City and plots them on a map. Maps help visualize regional patterns, distributions, and location-based insights.

g) Combine Multiple Charts into a Simple Interactive Dashboard

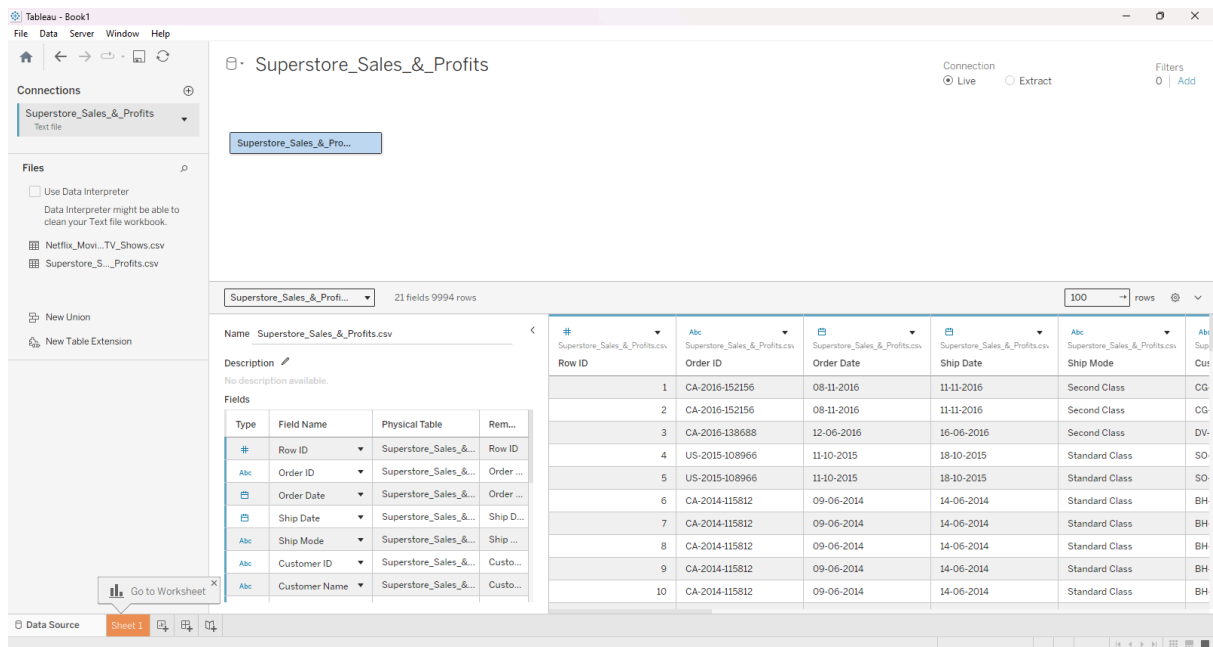
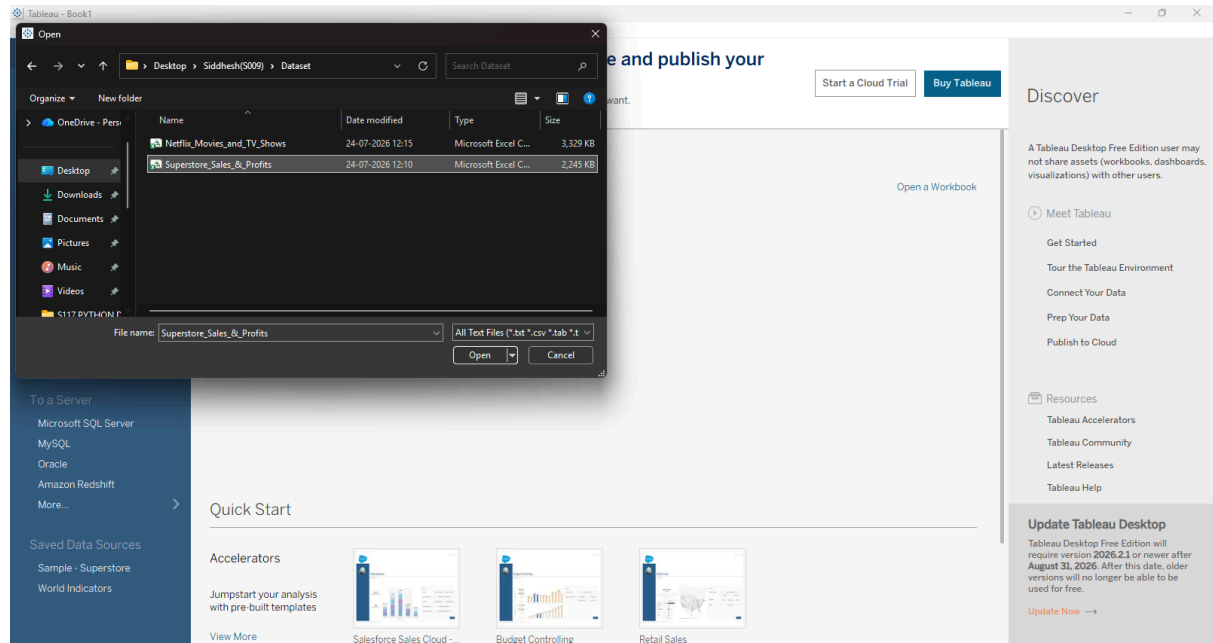
A dashboard combines multiple worksheets into a single interactive view. Filters, actions, and highlights can be added so users can explore and compare data across different visualizations in one place.

Part B – Program

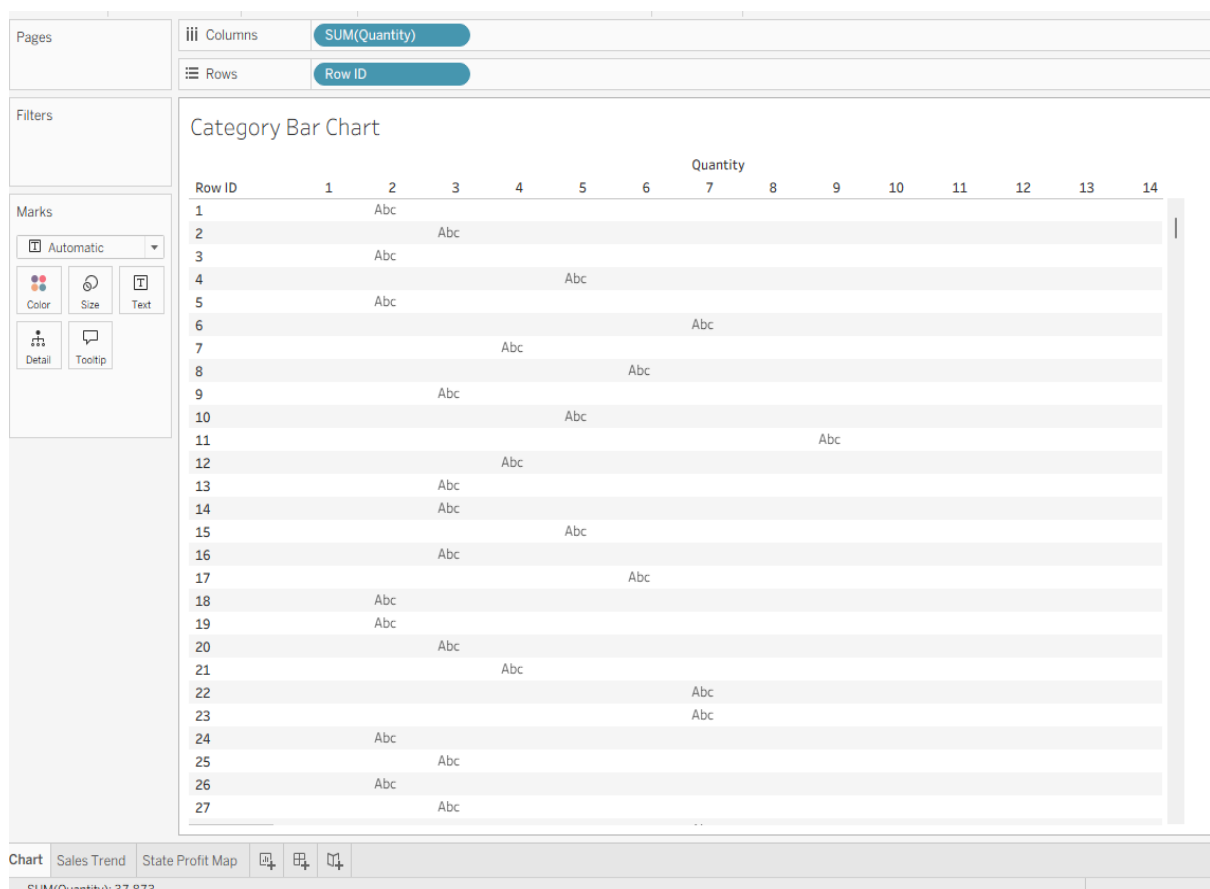
Getting Started with Tableau and Building Basic Visualizations

a) Connect to a CSV dataset and explore the Tableau interface

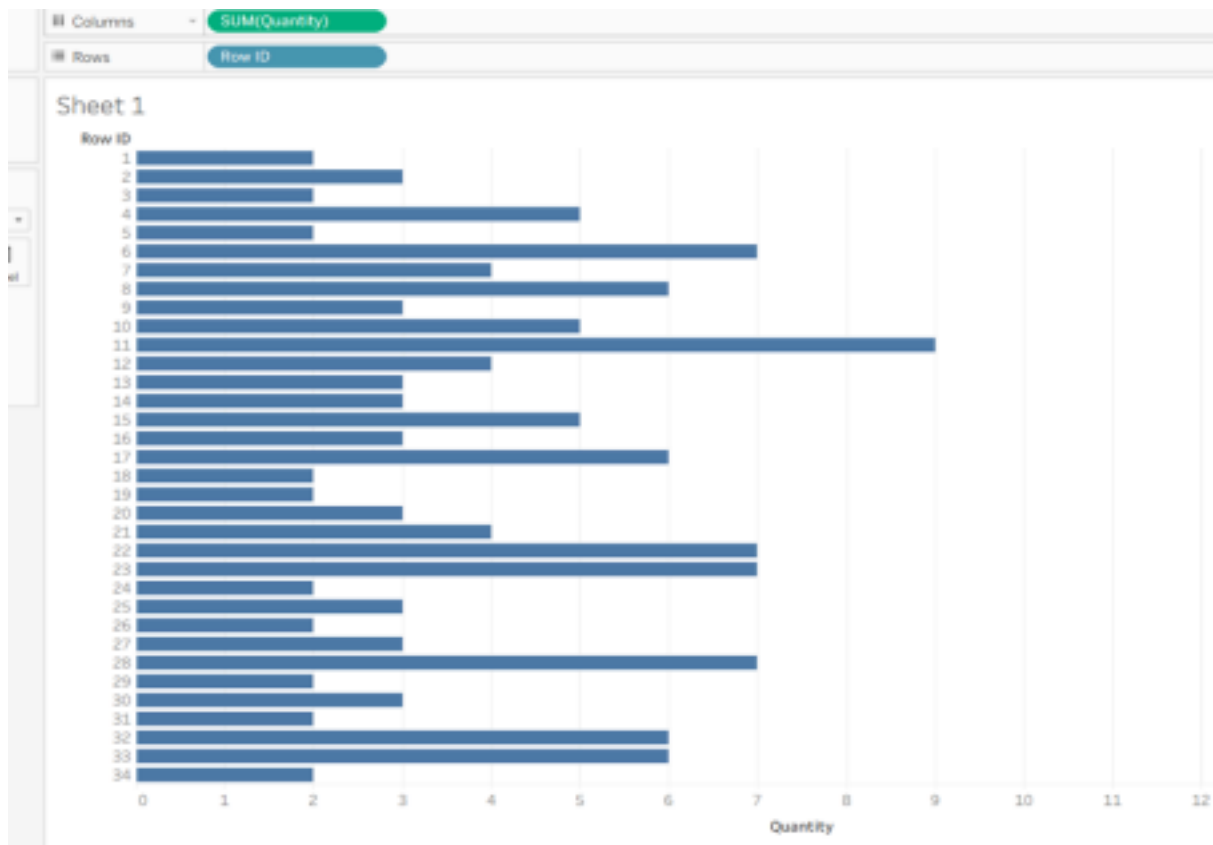
Output:



Output:

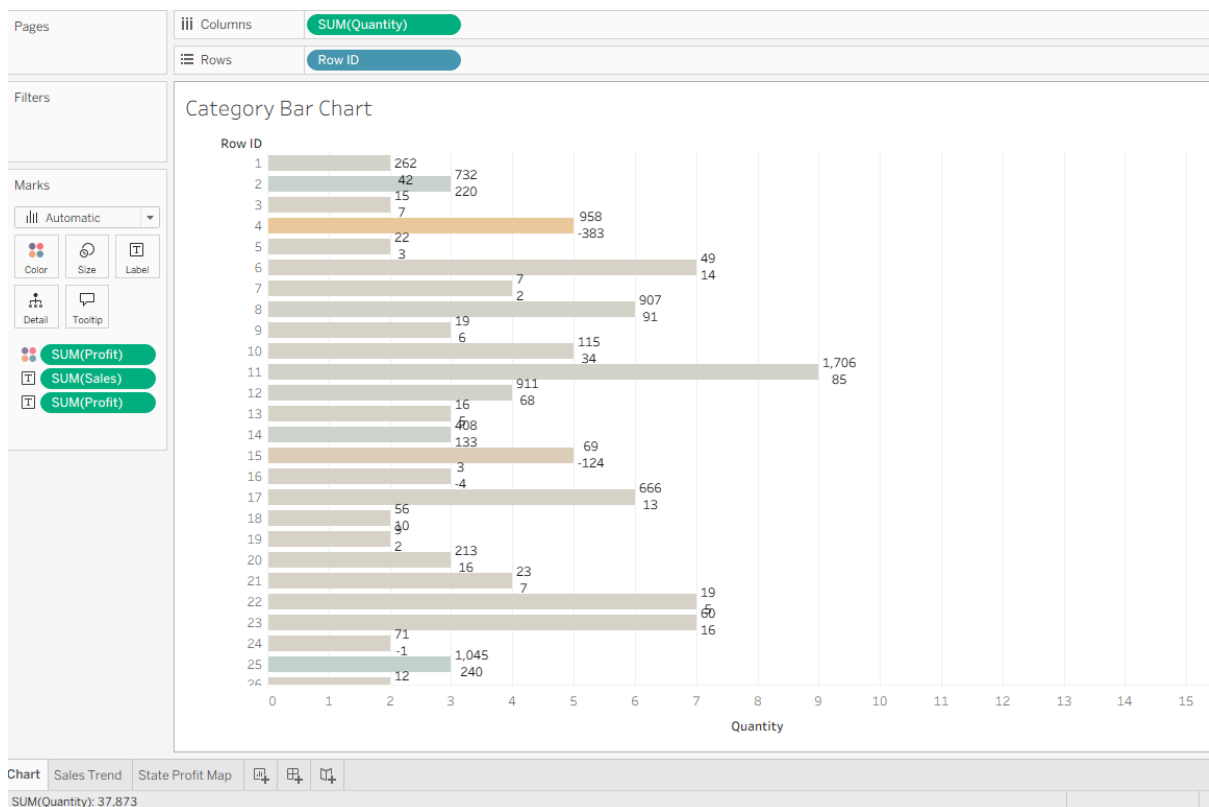


Continuous Output:



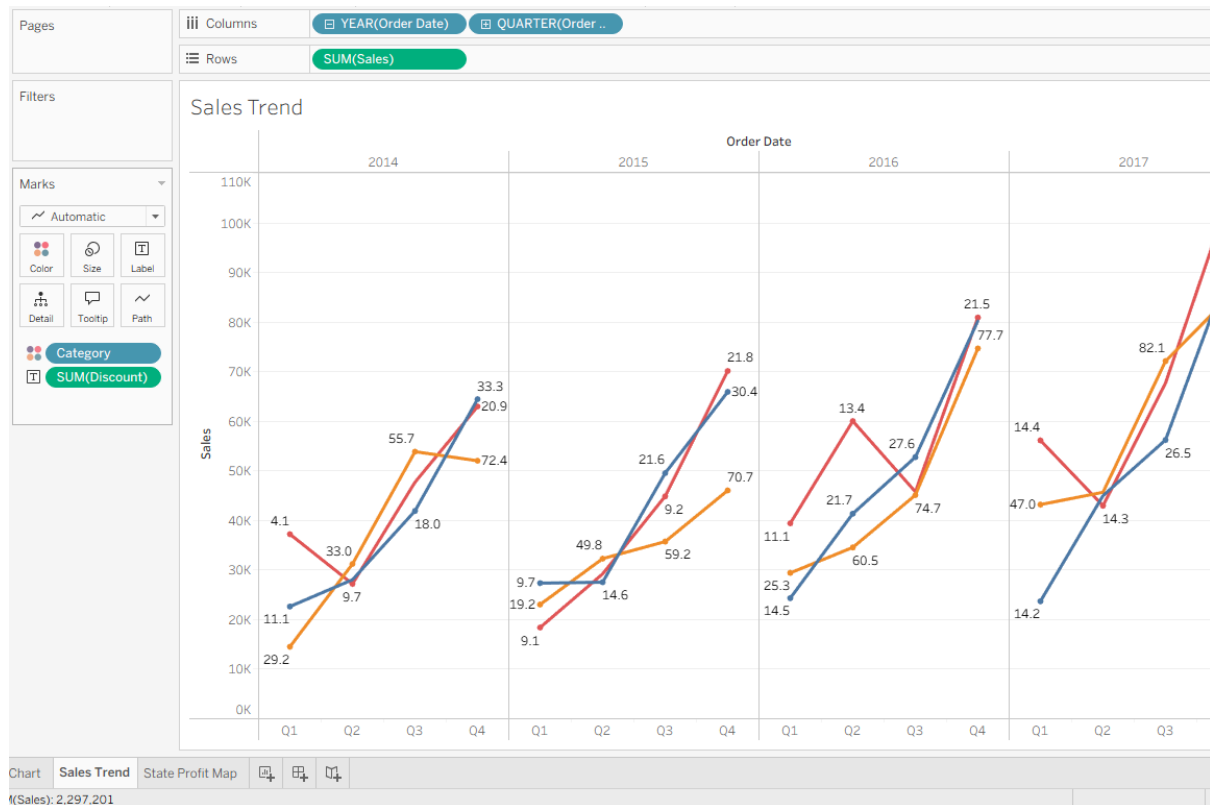
d) Create and customize a basic bar chart

Output:



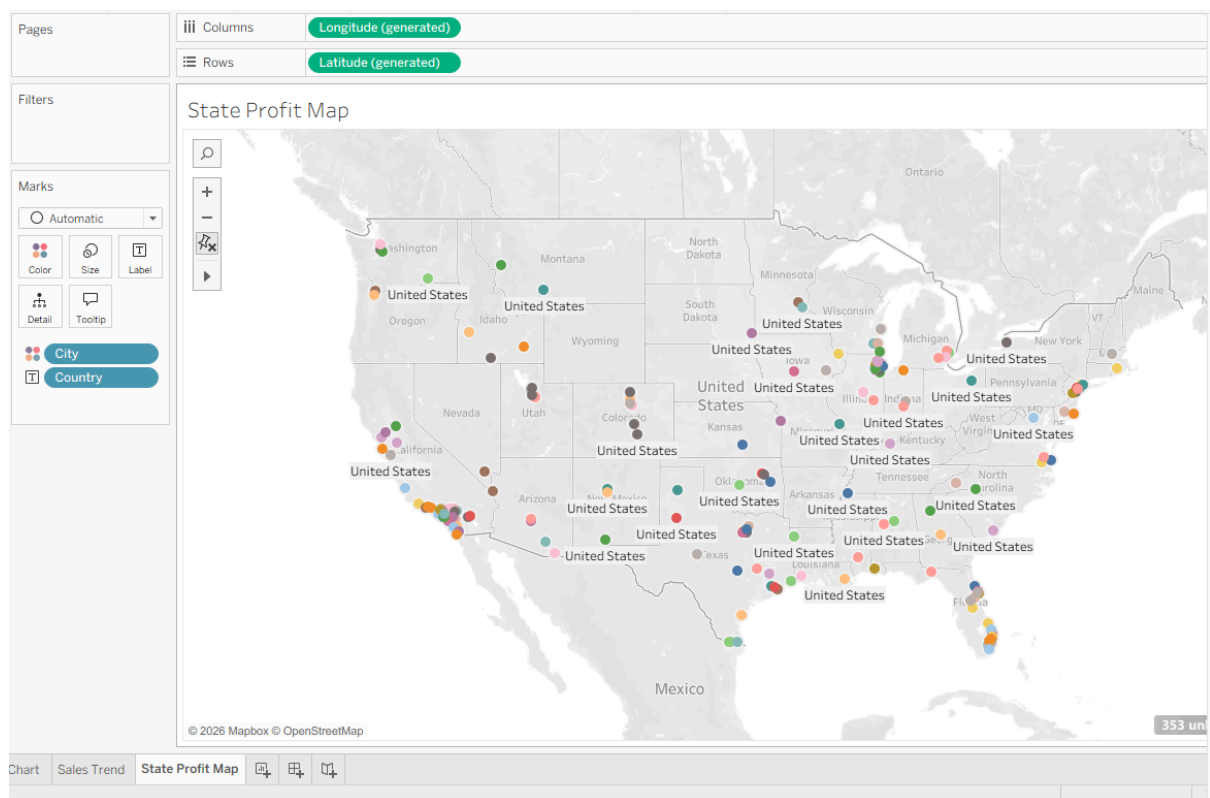
e) Build a line chart for time-series analysis

Output:



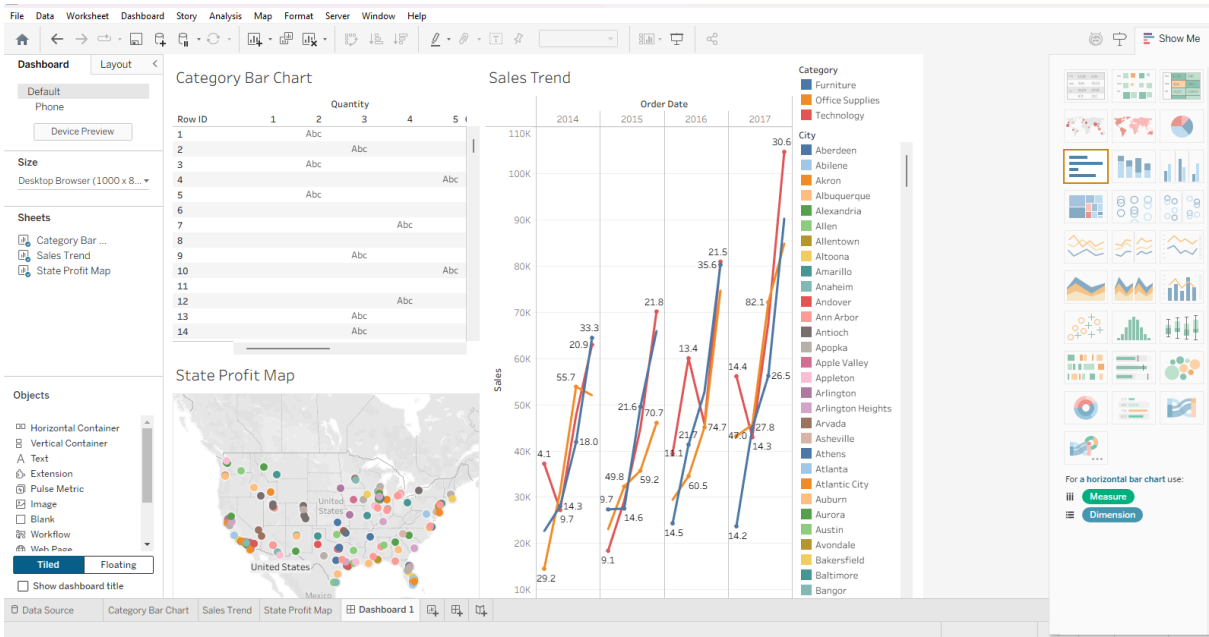
f) Create a geographic map visualization

Output:



g) Combine multiple charts into a simple interactive dashboard

Output:



Part C – Curiosity Question

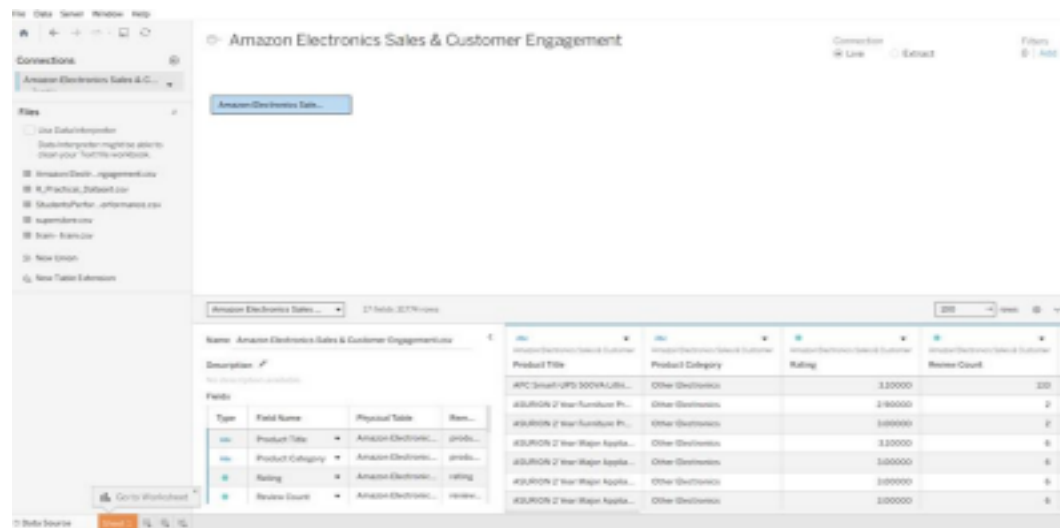
Curiosity Question

Question:

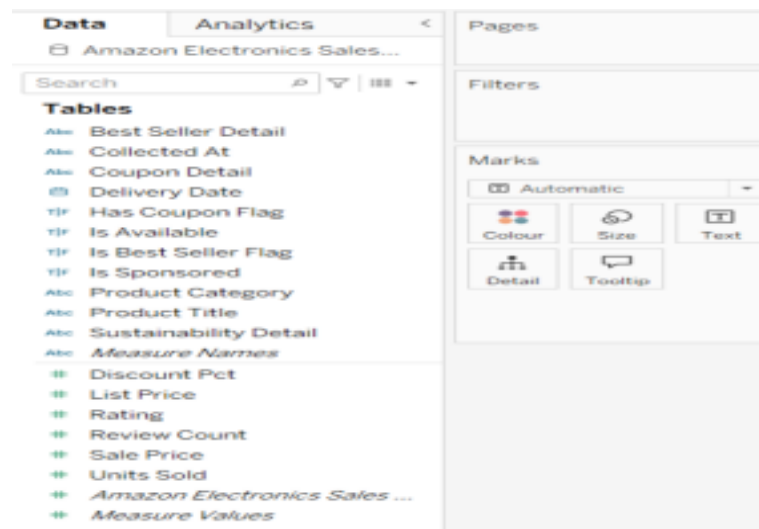
Using a dataset of your choice (CSV format), create an interactive Tableau dashboard that effectively presents key insights from the data.

Requirements:

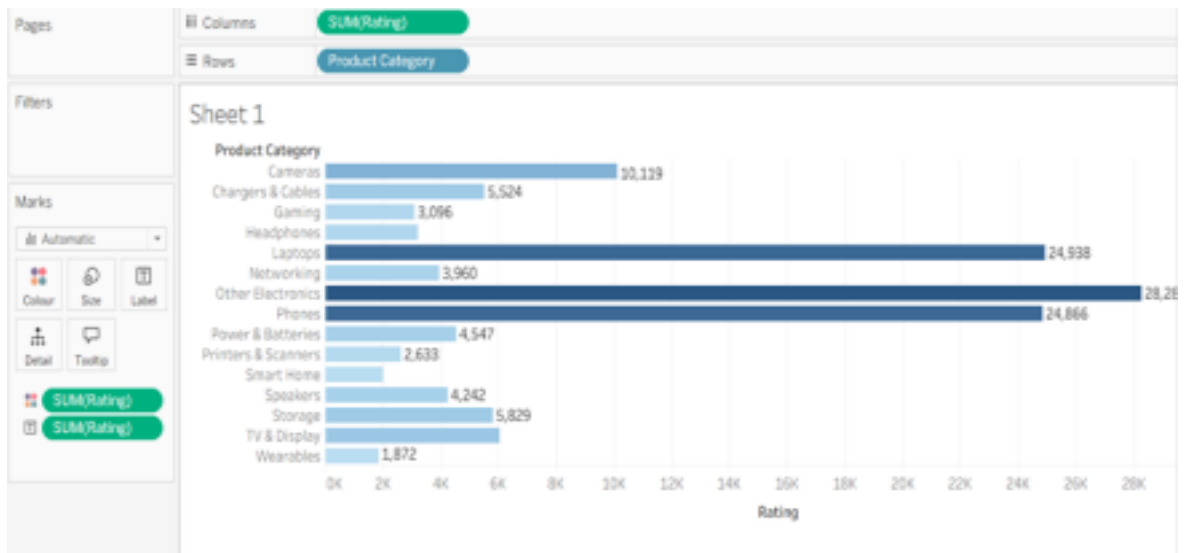
1. Import your own dataset into Tableau.



2. Identify appropriate **Dimensions** and **Measures**.



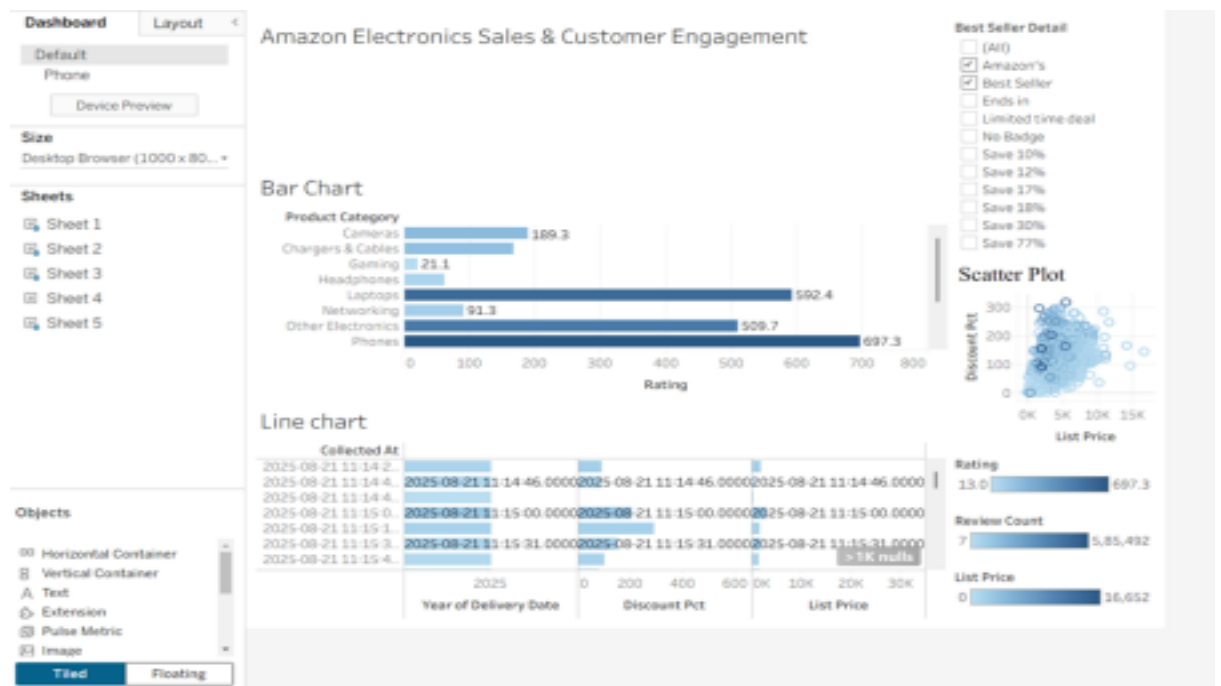
3. Create at least **three different visualizations** (e.g., bar chart, line chart, map, pie chart, scatter plot, etc.).



- Add at least **one filter** or interactive element to the dashboard.



- Arrange the charts into a **single interactive dashboard** with a clear title.



- Write **3–5 observations** explaining the insights gained from your dashboard.

- Phones and Laptops have the highest average ratings** among all product categories, indicating that customers are generally more satisfied with these products than with other categories.

- 2) **Gaming and Headphones have comparatively lower ratings**, suggesting these categories may require improvements in product quality, pricing, or customer experience.
- 3) **There is no strong correlation between List Price and Discount Percentage**, as products across different price ranges receive varying discounts. This indicates that discounts are not based solely on product price.
- 4) **Review Count varies significantly across products**, with a few products receiving a very large number of reviews. This suggests that some products are much more popular and attract higher customer engagement than others.
- 5) **The interactive filter allows users to analyze specific product groups**, such as Amazon's Best Sellers. Filtering the dashboard makes it easier to compare product performance and identify trends within selected categories.

Submission:

- Tableau Workbook (.twb or .twbx)
- Dataset (.csv)
- Screenshot of the completed dashboard
- A short report containing the observations.